

Slide 1: Title Slide

- Hello! My name is Nikhil Ghugare, and today I'll be presenting our project titled “**Interpretable Machine Learning for Youth Substance Use Analysis**”, using the **NSDUH 2023 dataset**.
 - Our goal was to use machine learning, specifically **decision trees**, to uncover what drives tobacco and substance use among teens.
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Slide 2: Why Study Youth Substance Use?

- Substance use at a young age is closely linked to long-term health and behavioral risks.
 - To reduce future harm, it's critical to understand the social and environmental factors that influence these behaviors.
 - Using interpretable models like decision trees allows us to find patterns that are easy to explain and communicate, especially to policymakers or school-based programs.
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Slide 3: Dataset Overview

- The data comes from the **National Survey on Drug Use and Health (NSDUH)**, filtered to include only youth under 18.
 - Sample size is around 10,500 youth responses.
 - We focused on three major variable types:
 - **Substance Use:** TOBFLAG, SMKLSMDAYS, IRCIGFM
 - **Demographics:** INCOME, YOGRPFT2, HEALTH2
 - **Social Influence:** PARHLPHW, SCHFELT, FRDMJMON
 - Key challenges included handling missing values, special coded responses, and class imbalances in the target variables.
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Slide 4: Methodology

- We created **engineered features** to better represent social context:
 - **SCHOOL_FAMILY** = Parental involvement × school satisfaction
 - **PEER_INFLUENCE** = Peer pressure × friend behavior
 - **Preprocessing** included mean imputation for missing values and SMOTE to address class imbalance.
 - **Models** used:
 - DecisionTreeClassifier for binary and multi-class classification
 - DecisionTreeRegressor for predicting numerical values
 - All models were tuned with max_depth=5 and min_samples_leaf=10 to balance interpretability and performance.
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Slide 5: TOBFLAG – Binary Classification

- **Objective:** To predict whether a youth has ever used tobacco based on behavioral, social, and demographic features.
- **Model:** Decision Tree Classifier
 - **Accuracy:** 72.8%
 - **Recall (users):** 64%
 - **Precision (users):** 19%
- The tree had 55 nodes, 28 leaves, and a maximum depth of 5.
- Most influential features:
 - **PARHLPHW:** parental help
 - **INCOME**
 - **PEER_INFLUENCE**
- The model performed reasonably well but showed some precision limitations due to class imbalance.

Plots Explanation

- The confusion matrix reveals the model performs well in predicting **non-users** (1417 true negatives), but has difficulty detecting **actual users**, with **122 true positives** and **70 missed cases**, along with **504 false positives**—indicating potential class imbalance.

- The **feature importance plot** shows **Peer Influence** as the strongest predictor of tobacco use, followed by **group risk behavior (YOGRPFT2)** and **self-reported health (HEALTH2)**. These social and behavioral indicators significantly influence whether youth are likely to use tobacco.
 - This highlights the importance of targeting peer environments and risk behavior in prevention efforts.
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Slide 6: SMKLSMDAYS – Multi-Class Classification

- Objective: Classify how many days smokeless tobacco was used (e.g., 0, 5, 10, etc.).
- Model: Decision Tree Classifier
 - Accuracy: 53.3%
 - Best F1-score observed in the majority class
- The tree had 33 nodes and 17 leaves.
- Key predictors:
 - HEALTH2: Self-rated health
 - PEER_INFLUENCE
 - INCOME
- The model provided good interpretability across the classes despite moderate performance.

Plot Explanation

- The **confusion matrix** reveals that the model is highly accurate in predicting frequent users (label 5), but performs poorly on lower usage categories (labels 1–4), with most samples getting misclassified into the highest class. This is a common issue due to class imbalance in real-world survey data.
- The **feature importance plot** shows that **self-reported health (HEALTH2)** and **income level** are the strongest predictors of smokeless tobacco frequency, followed by **peer influence**. Youth reporting poorer health or lower income were more likely to be frequent users.
- These insights can inform targeted interventions by health programs focusing on vulnerable groups.

Slide 7: IRCIGFM – Regression

- Task: Predict how many days in a month a youth smoked cigarettes.
 - Model: Decision Tree Regressor
 - R^2 Score: -0.36 (model underfits)
 - MAE: 4.97 days
 - RMSE: 7.00
 - Accuracy within ± 0.5 days: 16.7%
 - Top variables:
 - INCOME (importance score: 0.38)
 - PEER_INFLUENCE
 - SCHOOL_FAMILY
 - While performance was limited, it still highlighted some meaningful variable influences.
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Slide 8: Decision Path for Regression

Goal: The main objective was to estimate monthly smoking frequency based on social and personal factors.

Root Split: The first split in the tree is on peer influence. This tells us that peer behavior is the most powerful driver of youth smoking. Lower peer influence generally leads to fewer smoking days.

As we move down the tree:

- Low peer influence, combined with low income and a safe school environment, predicts around 2.2 days per month.
- On the other hand, high peer pressure, poor health, and weak family-school support can lead to smoking as much as 13.6 days per month.

Top predictors across the tree include:

- PEER_INFLUENCE
- INCOME
- HEALTH2 (health status)

- SCHOOL_FAMILY (a feature combining school safety and family help)
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Slide 9: How do Decision Tree Splits

- **First**, they're rule-based. Each split is based on a specific condition or threshold, for example, "Is income greater than \$40K?" If yes, go one way; if no, go the other. This makes the decision-making process structured and logical.
 - **Second**, they're human-readable. Unlike complex black-box models, decision trees give us clear logic that we can trace and explain. This is super helpful when we're working with sensitive topics like youth behavior, we want transparency.
 - **And third**, they're mix-friendly. Trees can handle both numerical data, like income, and categorical values, like school attendance or parental involvement, without any special encoding tricks.
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Slide 10: How Model Learns from Variables.

- Binary Inputs are simple yes/no flags, like whether a parent helps with homework. These are useful for understanding the presence or absence of support systems.
 - Ordinal Inputs include variables like self-reported health scores. These follow a natural order—say from "poor" to "excellent"—which helps the model assess well-being levels.
 - Continuous Inputs involve raw numeric values, such as the number of days a substance was used. These offer precise quantitative insight and allow finer-grained splits in the decision tree.
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Slide 11: Summary and Takeaways

- Binary Classification: Parental help, income, and peer influence are strong predictors of tobacco use.

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- Multi-Class Classification: Health and peer context were dominant for predicting frequency of usage.
- Regression: While performance was low, income and social factors emerged as important signals.
- Decision trees provided an interpretable and explainable way to extract value from complex survey data—making them ideal for public health analysis.